BME423 · Side 1 · describe and calculate probability

Through Session 13 · balanced ANOVA · two-sided tests unless specified

Identify the data scale before choosing a test

Ask: measured amount or category? If measured, continuous or discrete? If a category, ordered or unordered?

Mean, variance, SD, and SEM

```
x̄ = Σx/n; s² = Σ(x−x̄)²/(n−1); s = √s²; SEM = s/√n
```

For a short list, build columns x, x−x̄, and $(x-\bar{x})^2$. Add the last column once.

x̄: sample average; s^2: sample variance; s: sample SD.

If SD is provided, square it before using a variance formula. Do not divide individual-measurement probabilities by √n.

Median, range, percentiles, and IQR

```
R = p(n+1)/100; percentile = x[I] + fraction × (x[I+1]−x[I])
```

Find the two bracketing observations, then move the required fraction of their gap.

I: integer part of R, using first observation as rank 1.

Use this lecture convention by hand. In R, quantile(x, probs, type=6) matches it for interior ranks.

Check a discrete probability function

```
f(x) ≥ 0; Σf(x) = 1
```

The full validity check is two lines: all entries nonnegative; total equals one.

The discrete distribution function is a running total

```
F(x)=Σ f(y) for y≤x
```

Sort the possible values and make a running-total column.

Expected value: value times probability, then add

```
E[X]=Σ x f(x); E[h(X)]=Σ h(x)f(x)
```

Use columns x, f(x), x f(x), and x^2 f(x). They supply both mean and variance inputs.

Variance of a random variable: the shorter route

```
Var(X)=E[(X−μ)²]=E[X²]−μ²
```

Reuse the x f(x) and x^2 f(x) columns rather than calculating every squared deviation again.

Variance uses squared units; SD uses the original units.

Keep extra digits until the final subtraction.

Integrate a density one small step at a time

```
∫ x^k dx = x^(k+1)/(k+1), k≠−1; ∫[a to b] f = G(b)−G(a)
```

Integrate first; upper value minus lower value second. Keep those jobs on separate lines.

Continuous CDFs, tails, and medians

```
F(x)=∫[−∞ to x] f(u)du; P(a<X<b)=F(b)−F(a)
```

Find F once, then reuse subtraction, complement, or an inverse equation.

BME423 · Side 2 · Normal and sampling

Through Session 13 · balanced ANOVA · two-sided tests unless specified

Uniform distributions: probability is a length ratio

$P(c<X<d)=(d-c)/(b-a)$; $E[X]=(a+b)/2$; $Var(X)=(b-a)^2/12$

For a flat density, use rectangle widths instead of integrating from scratch.

Continuous expectations and variance

$E[X]=\int x\, f(x)dx$; $E[X^2]=\int x^2 f(x)dx$; $Var(X)=E[X^2]-\mu^2$

Work over the support only; write the new integrand before integrating.

Normal distributions: separate the measurement from its z score

$z=(x-\mu)/\sigma$; $x=\mu+z\sigma$

Write subtract, then divide. Do not evaluate the bell-shaped density formula for a table probability.

$\Phi(z)=P(Z\le z)$, the left-tail cumulative area.

Normal probabilities: left, right, or between

Left: $\Phi(z)$; right: $1-\Phi(z)$; between: $\Phi(z_{upper})-\Phi(z_{lower})$

Left = table; right = one minus table; between = upper table minus lower table.

Normal percentiles: turn area back into a measurement

$x_p = \mu + z_p\sigma$; qnorm(p, mean=μ, sd=σ)

Probability given → qnorm or inverse table → location. Location given → pnorm or table → probability.

Sampling distributions and the central limit theorem

$E[\bar{x}]=\mu$; $Var(\bar{x})=\sigma^2/n$; $SD(\bar{x})=\sigma/\sqrt{n}$

Circle average or mean in the question. That is your signal to use SEM in the denominator.

Rescaling and adding independent random variables

$E[aX+b]=aE[X]+b$; $Var(aX+b)=a^2Var(X)$; $Var(X-Y)=Var(X)+Var(Y)$ for independent X,Y

Means keep the signs; independent variances add with squared coefficients.

Unbiased estimates and sample size

$SEM=s/\sqrt{n}$; estimated $SE(s^2)=s^2\sqrt{[2/(n-1)]}$

If n is multiplied by k, SEM is divided by √k.

BME423 · Side 3 · means and comparisons

Through Session 13 · balanced ANOVA · two-sided tests unless specified

Choose the test from the question

Choose outcome → number of groups → study design → test → correction → conclusion.

The midterm review excludes Session 14 and sum-of-squares ANOVA; its ANOVA designs are balanced.

From a null hypothesis to a conclusion

Use a fixed answer frame: H0 → statistic → df → critical value or p → reject/fail to reject → result in context.

Balanced ANOVA: compare between with within

```
s²_within=average(s_i²); s²_between=n×variance(group means);
F=s²_between/s²_within
```

Square the supplied SDs → average them. Then take variance of the means → multiply by n → divide.

m = number of groups; n = subjects in each group.

Do not use the unequal-size sum-of-squares method for this midterm; the review explicitly says balanced ANOVA only.

Two independent means: pooled Student t

```
s_p²=[(n1−1)s1²+(n2−1)s2²]/(n1+n2−2); t=(x̄1−x̄2)/
√[s_p²(1/n1+1/n2)]
```

Think: pooled spread → uncertainty of the difference → difference divided by uncertainty.

R's default t.test uses Welch. Set var.equal=TRUE to reproduce the pooled procedure taught here.

One sample against a reference mean

```
t=(x̄−μ0)/(s/√n); df=n−1
```

Reference test = difference from reference ÷ SEM.

Pairwise comparisons after ANOVA

```
t_ij=|mean_i−mean_j|/√[s²_within(1/ni+1/nj)]; df=N−m
```

Equal n: reuse one denominator, $\sqrt{(2s^2_within/n)}$, for every pair.

Bonferroni: split α across the requested family

```
per-comparison α = overall α/k
```

Count the requested comparisons first, then adjust the level once.

Holm: strongest evidence first, then stop

```
rank j cutoff = α/(k−j+1)
```

Sort → compare → continue after rejection → stop at the first failure.

Ranking statistics directly requires the same reference distribution and degrees of freedom. Otherwise rank the p values.

BME423 · Side 4 · nominal outcomes, association, R

Through Session 13 · balanced ANOVA · two-sided tests unless specified

Binary outcomes: a proportion is a mean of zeros and ones

```
p̂=m/n; estimated Bernoulli variance=p̂(1−p̂); SE(p̂)=√[p̂(1−p̂)/n]
```

Label the event first. Death risk and survival risk use different numerators.

Two proportions: pool the event counts

```
p̂_pool=(m1+m2)/(n1+n2); SE0=√[p̂_pool(1−p̂_pool)(1/n1+1/n2)]
```

Add event counts and totals to pool. Do not simply average proportions when sample sizes differ.

Yates correction: reduce a discrete gap before testing

```
|z|=max[0,|p̂1−p̂2|−0.5(1/n1+1/n2)]/SE0
```

z: subtract half the reciprocal-size sum. χ^2: subtract half a count from each absolute cell gap.

Do not apply this course's 2×2 correction to a larger table or the multi-category goodness-of-fit example.

Expected counts in a contingency table

```
E_ij = row_total_i × column_total_j / grand_total
```

Row total × column total ÷ grand total, one cell at a time.

χ^2 contingency tests: turn cell gaps into one statistic

```
χ²=Σ (O−E)²/E; df=(r−1)(c−1)
```

Build columns O, E, gap, gap²/E. Add the last column. Use Yates gaps only for 2×2.

Goodness of fit: one set of counts against a proposed distribution

```
E_i=N p_i; χ²=Σ (O_i−E_i)²/E_i; df=k−1
```

One count list plus a target probability list → goodness of fit.

Study design, relative risk, and absolute risk

```
RR = risk_treatment/risk_control; ARR =
risk_control−risk_treatment
```

First write the two event fractions. Ratios describe relative risk; subtraction describes absolute change.

Use odds ratios rather than estimating disease risks from a case-control sample selected by disease status.

NNT and NNH: invert the absolute change

```
NNT=1/ARR; NNH=1/(risk_exposed−risk_control) for an adverse-event
increase
```

Subtract risks first, then take one divided by that difference.

Odds ratios: diagonal products with the table labeled

```
OR=(a×d)/(b×c), table [[a,b],[c,d]]
```

Label the table, multiply one diagonal, multiply the other, then divide.

Read R output without getting buried in it

```
pnorm(x,mean,sd); qnorm(p,mean,sd); summary(aov(y~group,data=d))
```

Read output in this order: test name → statistic → df → p → conclusion.